20th National High School Physics Contest

Final Problems

1.

5 equal-mass uniform spheres; set 4 of radius a at rest in a hemispherical bowl of radius R. Centers share a horizontal plane. A further 1 of radius b sits atop the 4. For what R value do the lower spheres separate? All contacts are smooth.

2.

An Earth satellite follows an ellipse with Earth at a focus. Its Cartesian equation is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

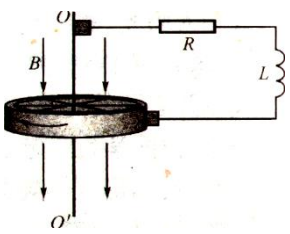
a、b are its known semi-major and semi-minor axes. At apogee, a large sudden energy input launches a probe of mass m forward; Earth's gravity gives a hyperbolic orbit.
Its focus is Earth. Its semi-transverse axis equals the original apogee distance. Find launch energy, given Earth mass m_E and Newton's constant G. Ignore other celestial bodies.

3.

The shown wheel has n spokes of length l and rotates about OO′ in a uniform magnetic field. Field is axial, flux density B. Brushes link axle and rim to R and L in a closed circuit. These are resistance and inductance; ignore metal flywheel/axle resistance and friction.

（1） A constant external torque M starts the wheel from rest. Find its steady angular speed Ω_0 and the current through resistance R, I_0。

（2） At steady speed remove torque M; speed Ω and current in R, I then vary with time. Different R values give different speed curves Ω（t） and current curves I（t）.
The supplied Ω（t） plots number 6. Take removal of M as the start: $t=0$,
Initially $\Omega=\Omega_0$ and $I=I_0$. The Ω and I plots each share a vertical scale;
all time scales agree. Choose physically possible angular-speed Ω versus-time curves Ω（t),
and the closest correct current I curve I（t）. Link each chosen Ω（t） with its matching I（t） curve by a straight solid line between their symbols below.
All links correct: full credit. Fewer correct links: partial credit. Any wrong link: zero.



Ω－1	I－A
Ω－2	I－B
Ω－3	I－C
Ω－4	I－D
Ω－5	I－E
Ω－6	I－F

4.

As shown, right of y axis the uniform field is $2B_0$; left of y axis it is B_0. Both fields